

Voice Recorder and Playback System



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There are several types of voice recorder and playback systems available in the market but most of them are expensive and their circuits are also very complex to assemble. Here is a simple circuit for recording and playback of voice messages. You can leave a voice message for your family or friends whenever you go out, which they can hear by pressing the 'play' button.

The circuit is built around a recording and playback chip that supports voice recording for 16 to 30 seconds and reproduces it clearly. It can be used in different types of applications such as door bells, railway announcement systems and automatic telephone answering devices.

Circuit and working

Fig. 1 shows the circuit of voice recorder and playback system. The circuit is built around voice recording and playback IC APR9301-V2 (IC1), voltage regulator 7806 (IC2), npn transistor BC547 (T1), 8-ohm, 0.5W speaker (LS1), electret microphone (MIC1) and a few other components.

IC APR9301-V2 is a high-quality voice recording and playback IC. The length of message recording depends on the value of external resistor R1 connected to its pin 7. The operation modes are described below.

Recording mode. When switch S1 is pressed, LED1 glows to indicate that recording has started. Now you can speak close to microphone MIC1 in order to record your message. You may

have to vary VR1 to adjust for different microphones. IC1 remains in recording mode as long as switch S1 is pressed and pin 27 of IC1 is grounded. Recording stops after 20 seconds (selected by 52-kilo-ohm resistance in this case), pin 25 of IC1 becomes 'high' and LED1 stops glowing.

The recording time duration can be increased or decreased depending on the value of resistor R1 as follows:

Test Points

Test point	Details
TP0	OV, GND
TP1	+6V
LED1	Glow when S1 pressed

PARTS LIST

Semiconductors:

- IC1 - APR9301-V2 voice recorder and playback IC
- T1 - BC547 npn transistor
- IC2 - 7806, 6V regulator
- LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 52-kilo-ohm
- R2 - 680-ohm
- R3 - 15-kilo-ohm
- R4, R5 - 10-kilo-ohm
- R6 - 100-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2, C6 - 0.01 μ F ceramic disk
- C3-C5 - 0.1 μ F ceramic disk
- C7 - 22 μ F, 16V electrolytic

Miscellaneous:

- LS1 - 8-ohm, 0.5W speaker
- MIC1 - Electret microphone
- CON1 - 9V DC connector
- S1 - Push-to-on switch
- S2 - 28-pin IC socket
- S1, S2 - 9V battery or 9V adaptor

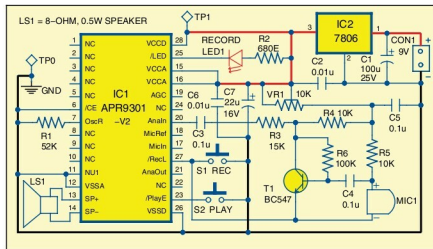


Fig. 1: Circuit of voice recorder and playback system

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